

Revision Report / Response to Reviewers

Initial Placement for Fruchterman–Reingold Force Model With Coordinate Newton Direction

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda
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Stephen Kobourov
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Dear Prof. Stephen Kobourov,

Thank you for directing us to the reviewers’ comments and for giving us the opportunity to revise our manuscript.

Based on the reports, we have carefully reviewed all comments and revised the manuscript accordingly. Our responses are provided below in a point-by-point manner. Where applicable, we indicate the location of changes in the revised manuscript by page and line numbers.

We hope the revised version is now suitable for publication and look forward to hearing from you.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda

A Response to Reviewer A

We thank Reviewer A for the careful and constructive assessment of our manuscript. We appreciate the reviewer’s observation that the proposed algorithm is interesting and that there are no major problems with the construction of the algorithm. We have revised the manuscript to address the concerns regarding the relationship between the proposed method and the Fruchterman–Reingold model, as well as the distinction between the FR and KK models.

A.1 Relationship Between the Proposed Approach and the FR Force Model

First, the authors do not sufficiently consider how well their approach matches the FR force model.

Originally, Fruchterman and Reingold [10] proposed the FR model and solved it by using a pseudo-physical simulation-based algorithm (by following Eades’ spring embedder method [8]). By contrast, Kamada and Kawai [21] proposed the KK force model and solved it by a Newton-like method. Also, many succeeding studies followed these approaches, i.e., either by combining simulation-based algorithms with the FR model or by combining Newton-like methods with the KK model. Although Hosobe [19] showed that both models could be solved with a Newton-like method called L-BFGS, the results still suggested that the KK model was more suitable for L-BFGS than the FR model.

However, this submitted paper adopts an unusual approach that combines a Newton-like method with the FR model. I understand that the paper is focused only on the

computation of initial layouts, but I think that the validation of adopting this approach is still insufficient.

Response: XXX

A.2 Results When Combined With FR and L-BFGS

It should be noted that the experimental results in Section 5 show that the proposed initial layout algorithm works better when it is combined with the L-BFGS algorithm for computing final layouts. It seems from Figure 6 that the combination of the proposed initial layout algorithm with the FR algorithm for computing final layouts does not work well except in limited cases like jagmesh1.

If the authors think that the proposed algorithm should be combined with the L-BFGS, they should adopt the KK model instead of the FR model, which would be more natural and more persuasive. If the authors think that the proposed algorithm works well for the FR layout algorithm (or another simulation-based algorithm), they should more deeply explore this direction and present clearer results in the paper.

Response: XXX

A.3 Difference Between FR and KK Models

Second, the authors do not sufficiently consider differences between the FR force model and others, especially the KK model, which becomes apparent especially in Subsection 7.1.

Here the authors first say “Although we utilized the coordinate Newton direction only for the optimization problem (1), we can see that its application is not necessarily limited to the FR force model alone”, and then present “objective functions arising from graphs”. Does the authors mean that this discussion applies to other force models including the KK model? If so, I think that the authors should adopt the KK model instead of the FR model.

Response: XXX

A.4 Graph Isomorphism Application

In addition, the authors present the graph isomorphism problem as an application of the proposed algorithm. I think that the KK model would be more suitable for this application than the FR model.

Response: XXX

A.5 Suggested Future Direction: Scalable FR-Based Methods

The following might work as hints for the authors’ paper revision. Intuitively, the FR model exhibits soft behavior while the KK model exhibits hard behavior. Therefore, the FR model is usually used for dynamic graph layouts like animation and interaction while

the KK model is usually used for static layouts. Also, the FR model is sometimes used for very large graphs because simulation-based algorithms for the FR model can be more scalable than Newton-like algorithms for the KK model. One plausible direction for the authors might be to combine the proposed algorithm with a scalable simulation-based algorithm for the FR model and to apply it to very large graphs.

Response: XXX

A.6 Notation: symbol E

- Symbol E is used for two different purposes, one to indicate a set of graph edges and the other to indicate energies between nodes possibly with subscripts and superscripts. It will be better to use different symbols for these different concepts.
- I wonder if $(i, j) \in V^2$ with $i \neq j$ in (1) is appropriately intended. This treats both (i, j) and (j, i) in the sum.
- Symbol ϵ appears only once in Subsection 4.4, which seems strange.
- Symbol f_i at line 6 of Algorithm 1 probably lacks the superscript “a”.
- In Subsection 5.3, the word “L-BFGS” that appears as superscripts of N does not seem to be correctly specified. The hyphen mark looks too long.
- In the optimization problem called “objective functions arising from graphs” in Subsection 7.1, “=” probably should be “:=”.
- The header of Subsection 8.1 seems to have a problem with the hyphen mark again.

Response: Thank you for pointing out these notation issues. We fixed them.

B Response to Reviewer B

We thank Reviewer B for the careful review and constructive suggestions. We appreciate the positive assessment that the paper addresses a topic of clear interest to the JGAA audience, that the technical explanation is comprehensive, and that the mathematical treatment is mostly easy to follow. We have revised the manuscript to improve the framing of the contribution, the discussion of related work, the experimental evaluation, and the organization of the later sections.

B.1 Related Work on Multilevel Approaches

In the Related Work section, in 3.1 Multilevel approaches are quickly dismissed. I don’t understand the statement the statement about sfdp being “prior work” to FR - as the latter came way later than the former. This sentence should be clarified. Also, I believe that further space could be given to multi-level approaches, also considering that could be one potential application/extension of this technique beyond this paper.

Response: XXX

B.2 Uncertain Framing of FR Versus L-BFGS

The framing of this paper is somewhat uncertain. Despite the paper title and general orientation towards FR, L-BFGS is consistently and repeatedly mentioned as a better alternative. Even in the related work section, in Section 3.1, only L-BFGS is mentioned as a ready-to-go measure for application in Multilevel approaches. In the experiments the paper explicitly mentions that FR is consistently outperformed by L-BFGS - then why we should use FR (even with the improved initial placement) at all? The paper does not make a good argument why we should choose that over the other, thus in a way making me question even the title of the paper. In general, it's unclear what's the impact of this research and how researchers could build upon it.

Response: XXX

B.3 Expansion of the Evaluation Dataset

The evaluation yields positive results, but in my opinion it could be further improved. First, I would suggest to expand 5.2 it in terms of number of graphs. More graphs from the Florida library could be used, or another source could be [1]. This expanded datasets already includes 3elt, but expands also on grids, cylinders, and other regular graphs that make up a solid benchmark set, and will provide a wider set of instances to discuss. If not all figures can be included in the main body of the paper, these could be moved in the supplemental material.

Response: XXX

B.4 Visualization and Color Map

Concerning the visualization of the layout, I don't believe that the index colouring adds much. A rainbow map, however, it's not a great choice. If color should be kept, maybe I would consider using a sequential continuous color map with a legend (see [2] for examples).

Response: XXX

B.5 Effect of Initial Placement Over Time

One thing that comes out of the charts in Figure 6 is that this different initial positioning has limited effect on the way the objective function decreases over time - which is an interesting phenomenon. In FR, the chart appears shifted below by how "better" the initial positioning is, with the non-CN version never catching up. Differently, it seems that the objective functions of L-BFGS and CN-L-BFGS tend to converge before the iteration cap. Does this mean that the positive effect of the initial placement tends to fade faster with L-BFGS, despite the better initial condition?

Response: XXX

B.6 Singularities in dwt_992 and collins_15NN

The dwt_992 and collins 15NN singularities could be described more. Would it be possible to generalize, highlighting what makes these two cases so different from each other?

Response: XXX

B.7 Spike in $f(X)$ near the initial time

Speaking of singularities, in the charts it seems like there is a spike in $f(X)$ right after the 0 seconds mark (like in 3elt, btree or cycle) is that an artifact?

Response: XXX

B.8 Organization of Sections 6 and 7

I am less persuaded from Sections 6 and 7. Section 6 includes considerations that I would include in Section 4, as to further justify your optimization design. I would also simplify the discussion, removing subsections and integrating their contents in Section 4. In Section 7 I would focus on the impact of this approach on graph drawing more specifically. For example, where is the next step for this technique, in the context of graph drawing? Will its presence in a drawing pipeline make a difference, or the gains in quality are not sufficient to justify its use? What about its inclusion in a multi-level pipeline (see also a similar point above)?

Response: XXX

B.9 Moving Section 8 to Supplementary Material

I would move Section 8 to supplemental material out of the main body of the paper.

Response: XXX

B.10 Bibliography

Bibliography must be updated. Some papers do not include any venue, others do not have a DOI, others only report their archive version (while journal versions are available).

Response: XXX

Closing

We again thank the editor and the reviewers for their valuable comments. We believe that the revised manuscript is clearer in scope, better positioned with respect to existing force-directed graph drawing methods, and more explicit about both the strengths and limitations of the proposed approach.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda